

Executive Summary

Why Slovene Refusal Outlasts English Abliteration

Internal Research Report — September 2026

Goal

GaMS3-12B-Instruct and Gemma-3-12B-IT share a pretrained ancestor (gemma-3-12b-pt) but diverge in safety supervision language: GaMS3 received Slovene continued pretraining and supervised fine-tuning with $\sim 7,000$ safety-refusal examples drawn from machine-translated Qwen-family responses, while Gemma-IT received Google’s standard multilingual post-training. Five iterations of experiments (23 artifacts total) tested whether Slovene-taught refusal transfers to English, whether English-only abliteration (Heretic) removes Slovene refusal equally in both models, and what mechanism explains any residual Slovene refusal.

Headline Findings

Finding	Source	Key number	Grade
Criterion shift (original models, HARD set)	Exp. 5 (iter. 2)	c DiD = $+0.52$ $[0.39, 0.67]$; d' DiD ≈ 0	ESTIMATE
C-OUT: output language drives residual refusal	Exp. 16 (iter. 5)	$\text{OUT}_{\text{Gemma}}(\text{PPI}) = +2.43$ $[+0.89, +5.38]$; $d\text{OUT} = -2.31$ $[-5.34, -0.68]$	ESTIMATE (WEAK)
R-BASE: lag is baseline, not edit-induced	Exp. 15 (iter. 4)	$G_{3,\text{edit}} = +0.47$ $[-0.96, 2.14]$; slopes ≈ 1	ESTIMATE
Identity anchoring	Exp. 5 (iter. 2)	FINAL DiD _{id} = 3.46 $[2.55, 4.62]$	PASS
Edit effective on RefusEU FINAL	Exp. 17 (iter. 5)	All 4 cells ROBUST ($n = 1,300$ per cell)	ROBUST
Utility: no SL damage	Exp. 12 (iter. 3)	All $H < 0.05$; excess asymmetry $+0.051$ $[0.012, 0.093]$	OK
R-INCAP: reserve is real safety	Eval. 4 (iter. 5)	Harmful-content OUT = 2.61 $[2.20, 3.07]$ at hi dose	ESTIMATE (WEAK)

Criterion shift on original models. Gemma over-refuses benign Slovene prompts (false-alarm rate 0.569 vs 0.364 in English), while GaMS3 is language-invariant (0.273 in both). The DiD in discrimination d' is near zero (-0.003); the DiD in criterion c is $+0.52$ $[0.38, 0.68]$. This replicates across experiments 5, 9, 14, and 16 on original models. On edited models the criterion-shift difference disappears (DiD_c = 0.14 $[-0.78, 1.12]$).

Output-language effect (C-OUT). Experiment 16 replicated on a fresh 294-item body with prediction-powered inference correction: edited Gemma refuses 2.43 log-odds more when told to answer in Slovene than in English. GaMS3 shows no output-language effect ($\text{OUT}_{\text{GaMS3}} = +0.12$ $[-0.47, +0.72]$). The input-language effect is negative in Gemma (-0.68 $[-1.23, -0.36]$): Slovene

input reduces refusal. The signal-detection decomposition places the effect in the criterion, not discrimination.

The lag is baseline (R-BASE). The cross-lingual ablation lag is better characterised as a pre-edit model difference than an edit-induced effect. On experiment 15’s fine-grained lambda ladder (MDE 0.42, the study’s most powerful estimate), the edit-induced component is $+0.47$ $[-0.96, 2.14]$ and both models’ curves run parallel with slopes near 1. In every item body, the pre-edit baseline $G_{3,\text{orig}}$ is negative.

Reserve is safety, not production failure. Gemma’s Slovene-output reserve produces less harmful content, not just more refusal: the harmful-content output-language contrast is 2.61 $[2.20, 3.07]$ at high dose. The four-class decomposition shows half of the non-compliance is deflection (production-failure index difference 0.46 $[0.23, 0.65]$), consistent with the relevance curse.

Dead Ends

Hypothesis	What killed it
Alt. 2: base-checkpoint geometry	delta- d' sign depends on layer; base geometry explains 4.2% of refusal variance
Alt. 3: persona gating	Manipulation check failed; premise (Slovene surplus) contradicted by the data
Alt. 4: prefill-depth signature	No readout can validly score prefill continuations ($\kappa \leq 0.21$)
Ancestor-direction induction	Pretrained direction never reached refusal rate 0.50 in any cell
Geometry hypothesis (M-a)	EN/SL directions nearly collinear at own L^* ($\cos \geq 0.986$); result is layer-dependent
Induction hypothesis (M-b)	Slovene-perpendicular component inert as refusal inducer (α_{50} right-censored)
Dose contrast (main vs alt. 1)	Underpowered across all 5 iterations; MDE exceeds $2m$ on every readout

What the Edit Does on RefusEU

On the full 1,300-item RefusEU FINAL reserved split (experiment 17), the Heretic edit at $\lambda = 1.0$ is effective in all four model $\times$ language cells:

Model	Language	ASR orig	ASR edit	Verdict
Gemma	EN	0.045	0.897	ROBUST
Gemma	SL	0.026	0.683	ROBUST
GaMS3	EN	0.033	0.970	ROBUST
GaMS3	SL	0.068	0.915	ROBUST

The edit is more effective on GaMS3 than Gemma in both languages. The English and Slovene prompt sets are different (LaBSE similarity 0.55), so within-model SL-vs-EN differences are not a pure language effect.

Utility Control

The English-only ablation edit does not damage Slovene downstream skills more than English ones. All macro headroom-normalised losses are below 0.05. Language asymmetry $A = H_{\text{SL}} - H_{\text{EN}}$ is positive (Slovene hurt less) in every cell. Random-adjusted excess asymmetry excludes zero for GaMS3 (+0.051 [+0.012, +0.093]). Localised costs: GaMS3 English ARC-Challenge (−3.7 pp, $p = 0.041$) and BoolQ (−2.0/−2.3 pp). Belebele comprehension and FLORES fluency are unaffected.

Evidence Quality Limitations

- **The readout gate has never passed.** Every LLM judge has kappa at most 0.43 on edited rows against blind adjudication; the paid gemini frontier judge has specificity 0.66–0.73 (over-calls REFUSE). All blind adjudication is author-LLM, not human.
- **The frozen-core Heretic procedure** (bf16, 80 trials, Heretic’s selection rule) has never been run. Every edit is an iteration-1 reduced-trial NF4 fallback.
- **Natural RefusEU reserved-split prompts** have never been scored under the edit (experiment 17 used the PolyGuard readout, not the pre-registered gemini judge).
- **R-BASE baseline is fragile** in most item bodies: experiment 15’s fragility index is 1 (one Gemma SL flip at $\lambda = 0$ moves the baseline inside the margin). Only experiment 9 has a robust offset (fragility index 9).
- **The harmful-content adjudicator kappa** is 0.12 in experiment 16 (gate requires 0.60), traced to truncation handling. Refusal kappa is 0.90.
- **GaMS3 failed the output-language manipulation** off the English-input row (SL→EN compliance 0.00 in experiment 14), so the input-vs-output decomposition is unavailable for GaMS3.

Novelty

The main claim is **PARTIALLY OCCUPIED** (downgraded from NOVEL). Two preprints partly occupy the lane: Addagada (2609.08373) forces the output language and finds a relevance curse; Nguyen et al. (2608.26186) cross prompt and response language on benign content. The narrowed design—crossing both factors with safety endpoints, a sibling pair sharing a pretrained ancestor, partial-dose edit plus a public checkpoint, Hungarian as a third language, and a constant suffix in all cells—remains unoccupied as of September 2026 (medium-low confidence).

Open Items for Future Work

1. C-OUT deserves pre-registered confirmation with a validated readout and human adjudication on edited rows.
2. If R-BASE is correct, the lag is an artefact of model selection (GaMS3 was trained to refuse less in Slovene), not a property of the ablation procedure.
3. The dose contrast (main claim vs alternate 1) remains underpowered across all five iterations; a larger item body or a different operationalisation of “dose” may be needed.
4. The M-OUT production-fluency account (refusal as a fallback when the model must produce harmful content in a language it generates poorly) was hypothesised but never directly tested.

Full report: 47 pages, 5 iterations, 23 artifacts, 32 tables, 7 figures. Spend: approximately \$25. All code and data artifacts are at github.com/ai-inventor-papers/ai-invention-2cf2a7-does-slovene-taught-refusal-survive.